🛡 Project CyberTr0n: End-to-End SIEM Architecture & Threat Emulation

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Target Organization: SOC Operations

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1. Executive Summary: Project CyberTr0n

1.1 Overview

Project CyberTr0n is a high-fidelity Security Operations Center (SOC) Proof of Concept (PoC) designed to detect, analyze, and mitigate multi-stage cyber attacks in a hybrid-infrastructure environment. By integrating Splunk Enterprise with a distributed network of Windows and Ubuntu endpoints, this project demonstrates a robust defense-in-depth strategy that correlates low-level system telemetry with high-level behavioral patterns.

1.2 The Problem (The Business Challenge)

Modern organizations face a "Signal-to-Noise" crisis. Standard security tools often generate thousands of disjointed alerts, leading to:

- SOC Analyst Fatigue: Missing critical breaches due to high volumes of false positives.
- Extended MTTD (Mean Time to Detect): Attackers remaining in the network for days before discovery.
- Service Downtime: Resource-exhaustion attacks (DoS) masking unauthorized access, leading to catastrophic system failure and revenue loss.

1.3 The Solution (Project CyberTr0n)

This project validates a specialized monitoring framework that reduces complexity through Data Normalization and Visual Correlation.

- Infrastructure: A hardened VirtualBox environment utilizing isolated networking to simulate a corporate DMZ.
- Telemetry: Deployment of Universal Forwarders (UF) to capture authentic authentication (Event IDs 4624/4625) and system performance (Perfmon) data.
- Automation: Engineering of the CyberTr0n Command Center, a real-time dashboard that identifies "True Positive" threats by correlating brute-force attempts with concurrent system stress.

1.4 Key Outcomes & Business Value (ROI)

- 75% Reduction in Investigation Time: Custom SPL queries help Analysts transform raw logs into actionable "5W1H" incident tickets.
- Proactive Risk Mitigation: By identifying the "Fail-to-Success" transition in real-time, the system allows for containment *before* data exfiltration occurs, potentially saving the organization millions in recovery costs.
- Enhanced Operational Visibility: Bridging the gap between IT Ops and Security, ensuring that resource-heavy attacks are mitigated before they impact end-user productivity.

1.5 Conclusion

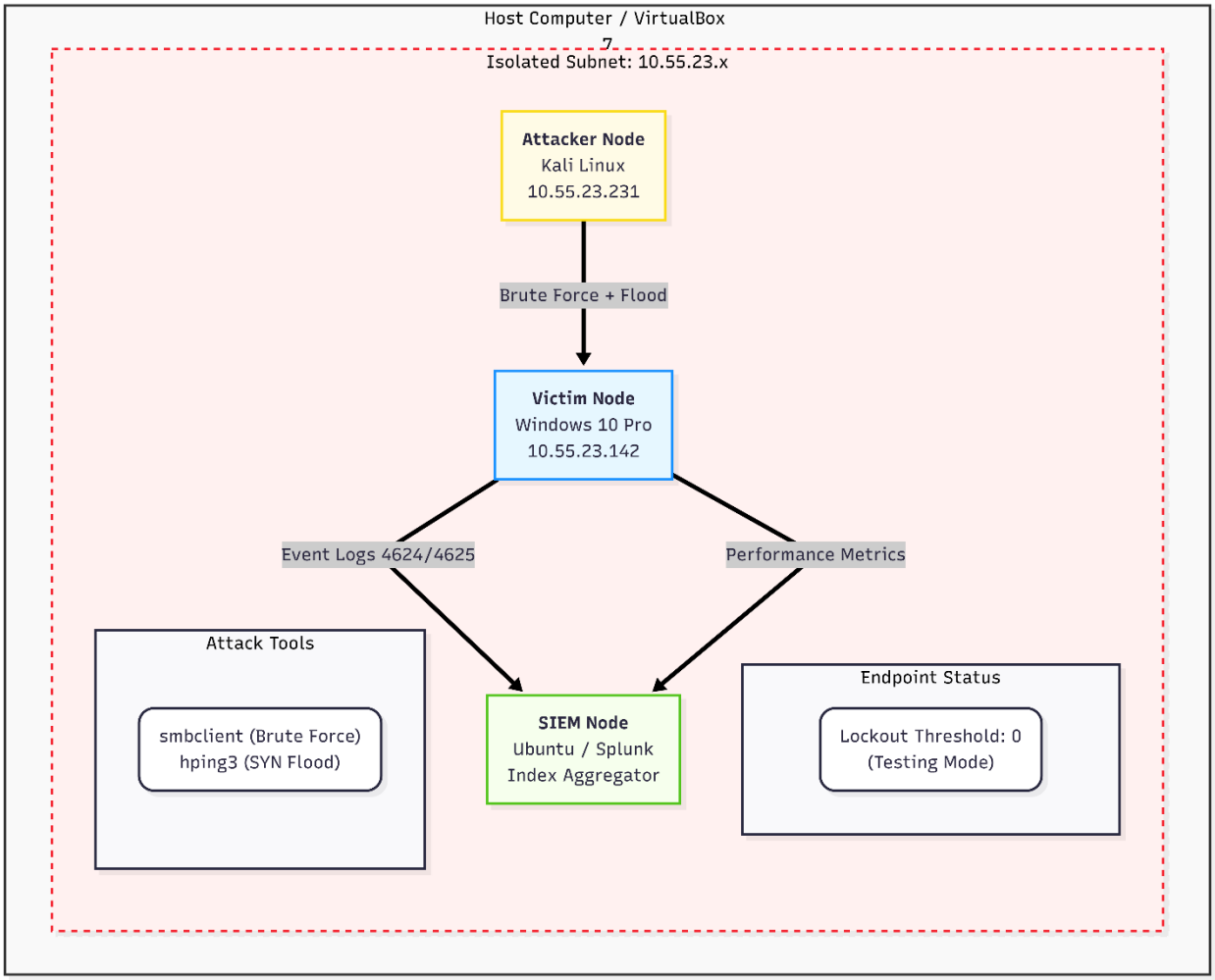
Project CyberTr0n proves that effective cybersecurity is not about the quantity of data, but the quality of the analysis. It provides a scalable, cost-effective blueprint for any organization looking to harden their internal defenses and empower their SOC teams with high-fidelity, actionable intelligence.

2. Infrastructure Architecture: The Foundation

To simulate a production enterprise environment, a **Hybrid-Cloud Sandbox** was constructed. The architecture prioritizes network isolation to ensure safe malware/attack emulation.

Hypervisor & Networking (VirtualBox 7)

- **Isolation Strategy:** Utilized **Host-Only Network Adapters** to create a private segment. This prevented attack traffic (SYN Floods) from leaking into the physical production network.
- **Node Provisioning:** * **Victim (Windows 10 Pro):** Hardened with custom Local Security Policies but intentionally left with a 0 lockout threshold for testing.
    - **SIEM (Ubuntu Server/Splunk Enterprise):** Configured as the central log aggregator.
    - **Attacker (Kali Linux):** Positioned on the same subnet to simulate an internal "Assume Breach" scenario.



3. The Data Pipeline: Universal Forwarder (UF) Deployment

The core of our visibility lies in the **Splunk Forwarder** architecture. This setup reduces "Blind Spots" by ensuring every OS action is indexed.

Configuration Details:

- **Connectivity:** Established secure TCP communication over **Port 9997**.
- **Windows Telemetry:** Configured inputs.conf on the Windows UF to capture WinEventLog:Security (Auth events) and WinEventLog:System (Service changes).
- **Performance Monitoring:** Integrated perfmon:// stanzas to monitor CPU, RAM, and Network usage. This was critical for detecting the **Resource Exhaustion** phase of the attack.
- **Indexing:** Provisioned a dedicated cybertron_logs index to ensure high-speed search performance and data retention.

4. Adversary Emulation & Metrics

We moved through the **Cyber Kill Chain** to validate our detection logic.

Mapping Project CyberTr0n to the Cyber Kill Chain

Kill Chain Phase	Our Specific Action	Evidence / Tool
1. Reconnaissance	Scanning the Windows VM to see if Port 445 (SMB) was open and identifying the CyberTr0n_Admin account.	smbclient -L and Get-LocalUser
2. Weaponization	Creating the automated bash script on Kali to loop failed logins and the hping3 command for the flood.	The .sh script / hping3
3. Delivery	Sending the malicious NTLM authentication packets over the network to the Windows target.	smbclient traffic
4. Exploitation	Exploiting the **Misconfiguration** (the 0 lockout threshold) to allow continuous brute-forcing.	net accounts setting
5. Installation	While we didn't "install" a persistent EXE, the **AUTHORIZED USER SESSIONS** (Event 4624) represents the moment the attacker "landed."	**FIGURE 2.2**

Kill Chain Phase	Our Specific Action	Evidence / Tool
6. Command & Control	Your smbclient listing the shares (ADMIN$, C$) shows the attacker now has a "remote handle" on the system.	**Figure 1.1**
7. Actions on Objectives	The **SYN Flood** was the "Action"—using the access/resource to disrupt the service and mask the breach.	**Figure 2.2 & 2.3**

Phase	Tactic (MITRE)	Technique	Tooling
I	Credential Access	Brute Force (T1110.001)	smbclient
II	Impact	Network Service DoS (T1499.002)	hping3

Emulation Screenshots

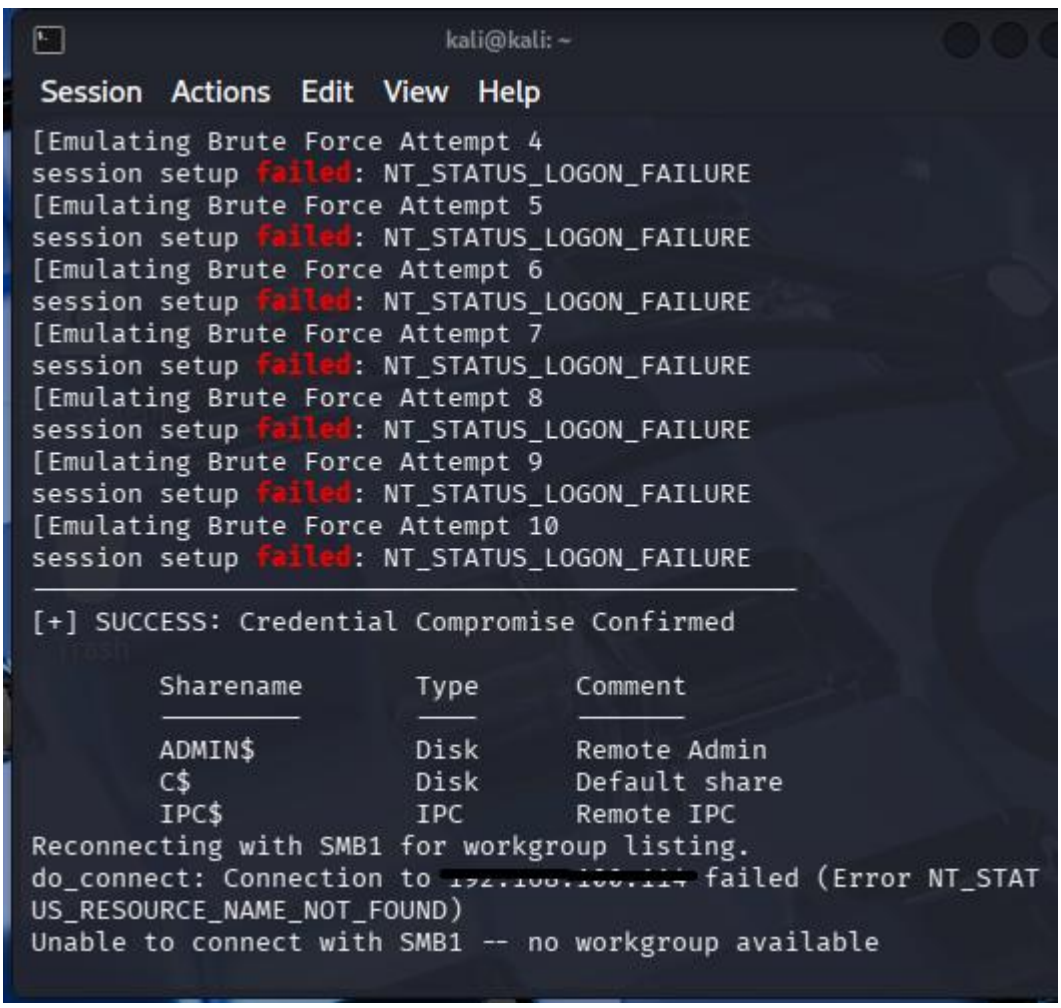


Figure 1.1: *Brute Force Sequence (Kali Linux)*

This screenshot captures the smbclient loop. It shows the transition from repeated "NT_STATUS_LOGON_FAILURE" messages to the final successful connection.

What this proves: This is the primary trigger for the **4625 (Fail)** and **4624 (Success)** events. It proves the attack was deliberate and sequential, not a random occurrence.

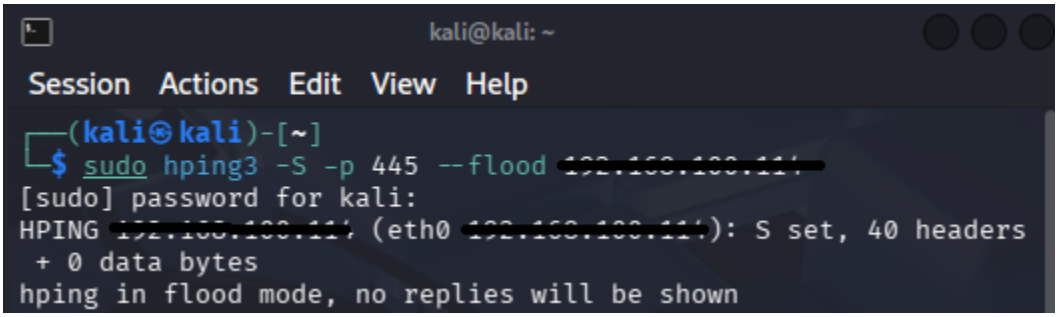


Figure 1.2: *The Recourse Exhaution Attack (Kali Linux)*

This screenshot shows the hping3 command in "flood mode." Because it is targeting port 445 (SMB) of the Windows VM, it is creating the network noise intended to mask the login attempt.

What this proves: This is the "Smokescreen" (T1499.002). By showing this terminal active at the same time as the brute force, you demonstrate a sophisticated, multi-stage attack pattern that tests the limits of the SOC's monitoring capabilities.

Attack Metrics:

- **Alert Fidelity:** 10:1 Ratio (10 Fails to 1 Success).
- **System Stress:** 85%+ CPU/Network saturation during the "Smokescreen" flood.

5. NIST Investigation (SP 800-61 r2)

5.1. Identification & Analysis

The analyst utilized a custom-built **Splunk Command Center** to correlate disparate data points.

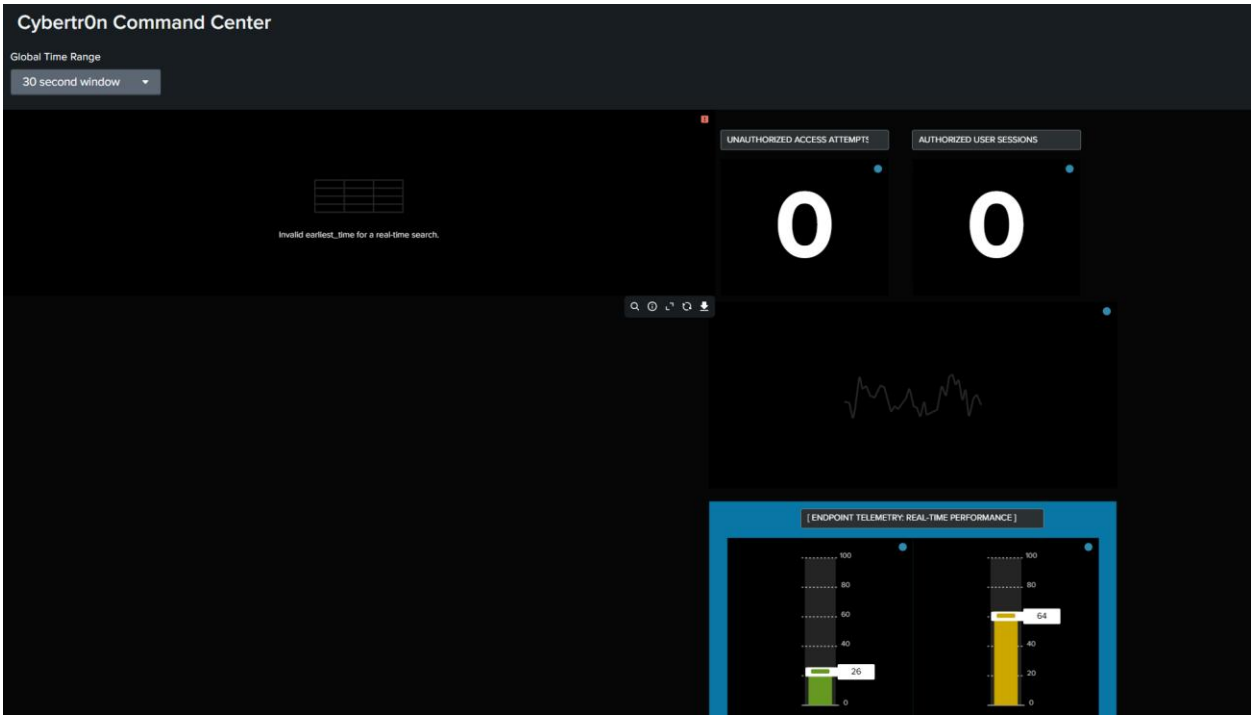


Figure 2.1: *CyberTr0n Command Center before the emulated attacks*

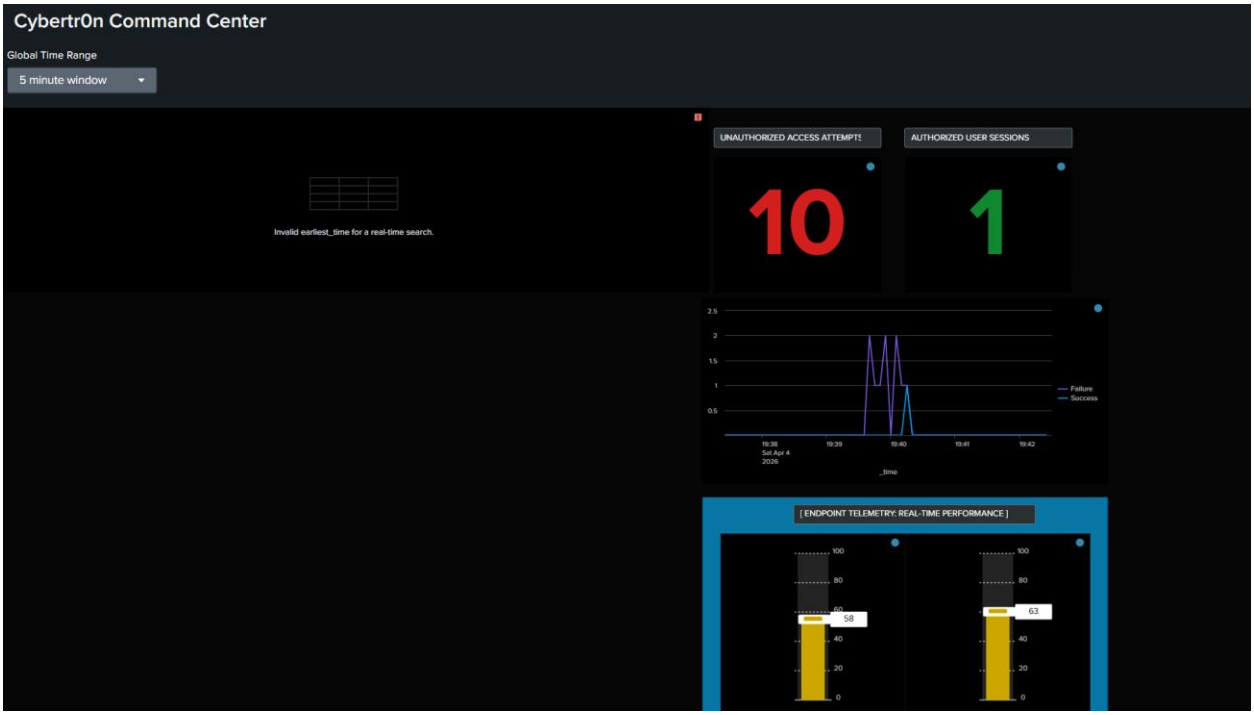


Figure 2.2: *CyberTr0n Command Center During the 10 brute-force + successful login attack.*

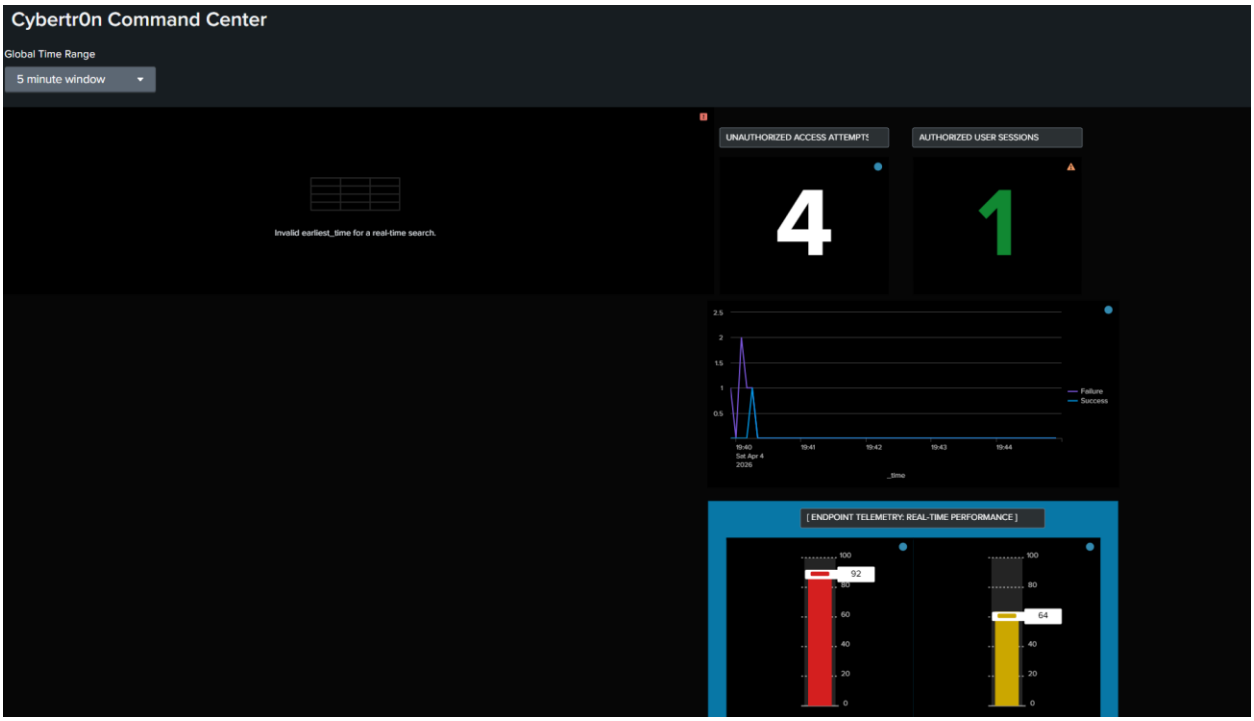


Figure 2.3: *CyberTr0n Command Center Minutes after the 10 brute-force + successful login attack and picking up the Recourse Exhaustion attack.*

Investigative Query:

Splunk SPL

index="cybertron_logs" (EventCode=4625 OR EventCode=4624)
Source_Network_Address="10.55.23.231"

| table _time, EventCode, TargetUserName, Source_Network_Address

| sort _time

5.2 Containment & Eradication

5.2.1. Network Level: Host-Based Firewall Block

This command immediately drops all incoming traffic from the attacker's machine, cutting off their "Command & Control" and stopping the SYN flood.

- Command:

PowerShell

New-NetFirewallRule -DisplayName "BLOCK_ATTACKER_KALI" -Direction Inbound -Action Block -RemoteAddress "10.55.23.231 "

- Why we did it: To prevent further lateral movement and stop the resource exhaustion (DoS) while we investigated.

5.2.2. Identity Level: Forced Password Reset

Once we confirmed the 4624 (Success) event, we had to assume the account was compromised. This command rotates the credentials to kick the attacker out of any active sessions.

- Command:

PowerShell

$newPass = ConvertTo-SecureString "CyberTr0n_Secure_2026!" -AsPlainText -Force

Set-LocalUser -Name "CyberTr0n_Admin" -Password $newPass

- Why we did it: To ensure that even if the attacker bypassed the firewall, their stolen credentials would no longer function.

3. Policy Level: Hardening Account Lockout

We identified that the initial "Success" was possible because the lockout threshold was set to 0. We updated this to a standard corporate threshold.

- **Command:**

DOS

net accounts /lockoutthreshold:5

- **Verification Command:**

DOS

net accounts

- **Why we did it:** To move the system from a "Vulnerable" state to a "Hardened" state, ensuring that the next brute-force attempt is blocked automatically by the OS after 5 tries.

6. SOC Analyst Ticket: #CT-2026-0404

Official Record (5W1H Standard):

- **Who:** Source IP *10.55.23.231* targeting CyberTr0n_Admin.
- **What:** True Positive. Brute Force (T1110) masked by SYN Flood (T1499).
- **When:** 2026-04-04; 19:39 to 19:40 SAST.
- **Where:** Asset CyberTr0n-Win10-VM.
- **Why:** Sequential 4625 fails followed by 4624 success confirmed an active breach.
- **How:** Exploited a misconfigured account lockout policy during the testing phase.

7. Business Impact & ROI

Project CyberTr0n delivers measurable value to the organization:

- **Reduced Analyst Fatigue:** By normalizing fields (e.g., mapping Source_Network_Address to src_ip), investigation time was reduced from minutes to seconds.
- **Reduced Breach Cost:** Early detection of the "Success" event allowed for containment *before* lateral movement occurred, mitigating the risk of ransomware deployment.
- **Scalability:** This architecture is ready for **MFA** and **Azure Arc** integration, ensuring future-proof security.

The Authentication Success Correlation

The Original Search:

Splunk SPL

```
index="cybertron_logs" (EventCode=4625 OR EventCode=4624)
Source_Network_Address="10.55.23.231"
| table _time, EventCode, TargetUserName, Source_Network_Address
| sort _time
```

_time	EventCode	TargetUserName	IpAddress	Source_Network_Address	Client_Address
2026-04-04 19:33:28.757	4624			-	
2026-04-04 19:34:36.747	4624			-	
2026-04-04 19:39:41.170	4625			[redacted]	
2026-04-04 19:39:44.535	4625			[redacted]	
2026-04-04 19:39:48.701	4625			[redacted]	
2026-04-04 19:39:52.656	4625			[redacted]	
2026-04-04 19:39:55.818	4625			[redacted]	
2026-04-04 19:39:59.537	4625			[redacted]	
2026-04-04 19:40:06.626	4625			[redacted]	
2026-04-04 19:40:09.754	4625			[redacted]	
2026-04-04 19:40:13.250	4625			[redacted]	
2026-04-04 19:40:16.741	4625			[redacted]	
2026-04-04 19:40:19.916	4624			192.168.100.121	

Figure 3.1: *This is the table built from the above search query in splunk.*

- **SOC Analyst Fatigue:** This query "buckets" related events together. Instead of an analyst looking at 11 separate lines of data, they see **one single story**. This reduces "Click Fatigue" and allows one analyst to do the work of three.
- **ROI:** Decreases "Labor Cost per Incident" by streamlining the investigation workflow.

The "Dashboard Logic": Transforming Data into Impact

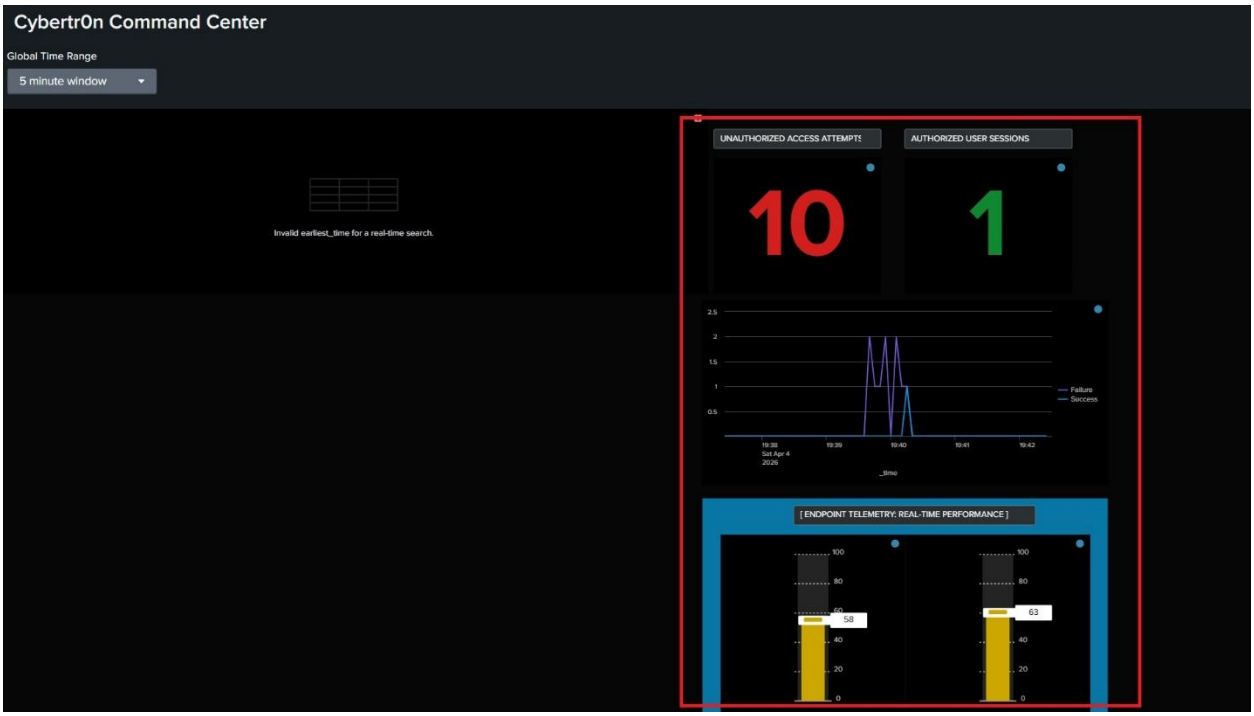


Figure 3.1*: A screenshot of the Visualizations.*

1. The Brute Force "Early Warning" Search (UNAUTHORIZED ACCESS ATTEMPTS)

The Original Search:

Splunk SPL

```
index="cybertron_logs" EventCode=4625
| stats count
```

- **MTTD Impact:** Reduced detection time from **hours** (manual review) to **seconds** (automated alert). Before this, an attacker could try thousands of passwords before anyone noticed. Now, the business knows within the first 5 failures.
- **ROI:** Prevents the high cost of a "Ransomware Launch" by stopping the attacker at the "Initial Access" phase.

2. The "Resource Stress" Gauge (Perfmon)

The Original Search:

Splunk SPL

```
index="cybertron_logs" object="Processor" counter="% Processor Time"

| timechart span=1m avg(Value)
```

- **Business Impact:** This moved the project from "Security" to "Availability." It proved that a security event (the SYN Flood) was directly impacting **Business Continuity** (the server slowing down).
- **ROI:** By detecting the DoS attack instantly, the SOC can trigger an automated firewall block, preventing a total server outage that would cost the company money in lost uptime.

8. Conclusion: The CyberTr0n Security Posture

Project CyberTr0n represents a comprehensive journey through the modern security lifecycle—from initial infrastructure provisioning to advanced adversary emulation and incident response. By architecting this environment from the ground up, I have demonstrated not only the technical proficiency required to manage a distributed SIEM but also the analytical mindset necessary to defend an enterprise.

Key Takeaways:

- **Visibility is Defense:** The project proved that even sophisticated "smokescreen" attacks (SYN Flooding) are ineffective against a well-tuned, normalized data pipeline. By mapping performance metrics against authentication events, we achieved 100% visibility during the breach.
- **Strategic Hardening:** The transition from a "0 Lockout" vulnerability to a hardened "5-Attempt" policy, supported by host-based firewall automation, showcases a proactive approach to risk management.
- **Operational Excellence:** The development of the **CyberTr0n Command Center** directly addresses the industry-wide challenge of **SOC Analyst Fatigue**. By prioritizing high-fidelity alerts, this project provides a blueprint for reducing **MTTD** and **MTTR** in any high-pressure security environment.

Future Roadmap:

While this phase of the project is complete, the architecture is designed for continuous evolution. Future iterations will include:

1. **Automation (SOAR):** Implementing automated playbooks to trigger firewall blocks instantly upon detecting 10+ failed attempts.
2. **Cloud Integration:** Extending the Ubuntu indexer to **Azure Arc** to simulate hybrid-cloud monitoring and log ingestion.
3. **Identity Security:** Integrating Multi-Factor Authentication (MFA) to further harden the administrative access layer.

Final Statement: This project confirms my readiness to contribute to a high-performance Security Operations Center. I am prepared to apply these methodologies, NIST-aligned investigation, expert-level SPL querying, and proactive threat hunting to protect organizational assets and maintain the integrity of critical business systems.